Question 18 — 1 mark

The range of $y = f(x)$ is $-6 \le y \le 12$. The range of the transformed function $y = af(x)$ is $-2 \le y \le 4$. Identify the value of a.

a) -3

b) $-\frac{1}{3}$

c) $\frac{1}{3}$

d) 3

Question 19 — 1 mark

Identify the expression which is equivalent to $3\log y - \frac{1}{2}\log x + \log z$.

a) $\log\left(\frac{y^3}{\sqrt{x}z}\right)$

b) $\log\left(\frac{y^3 z}{\sqrt{x}}\right)$

c) $\log\left(\frac{y^3}{x^2 z}\right)$

d) $\log\left(\frac{y^3 z}{x^2}\right)$

Question 20 — 1 mark

Identify the measure of the angle $-\frac{2\pi}{9}$ in degrees.

a) $-400°$

b) $-40°$

c) $40°$

d) $320°$

Question 21 — 1 mark

If $y = f(x)$ has a domain of $[2,5]$ and a range of $[6,10]$, identify the domain of $y = f^{-1}(x)$.

a) $\left[\frac{1}{2}, \frac{1}{5}\right]$

b) $[-5,-2]$

c) $[-10,-6]$

d) $[6,10]$

Question 22 — 1 mark

Identify which of the following is a polynomial function.

a) $p(x) = -\frac{1}{2}(x+2)^3(x-3)$

b) $p(x) = 2x^{\frac{1}{2}} + x - 3$

c) $p(x) = 3x^{-4} + x^2 - 6$

d) $p(x) = 2^x + 3$

Question 23 — 1 mark

Identify the total number of terms in the expansion of $(x-y)^9$.

a) 8

b) 9

c) 10

d) 11

Question 24 1 mark

Identify the exact value of $2\cos^2(15°)-1$.

a) 1

b) $\frac{1}{2}$

c) $\frac{\sqrt{3}}{2}$

d) $\sqrt{3}$

Question 25 1 mark

The zeros of the function $y=f(x)$ are $x=-2$ and $x=3$. Identify the zeros of the function $g(x)=2f(x-4)$.

a) $x=-6$ and $x=-1$

b) $x=2$ and $x=7$

c) $x=-4$ and $x=6$

d) $x=0$ and $x=10$

Question 26 1 mark

Identify the value of $\log_4\left(\frac{1}{4}\right)$.

a) -16

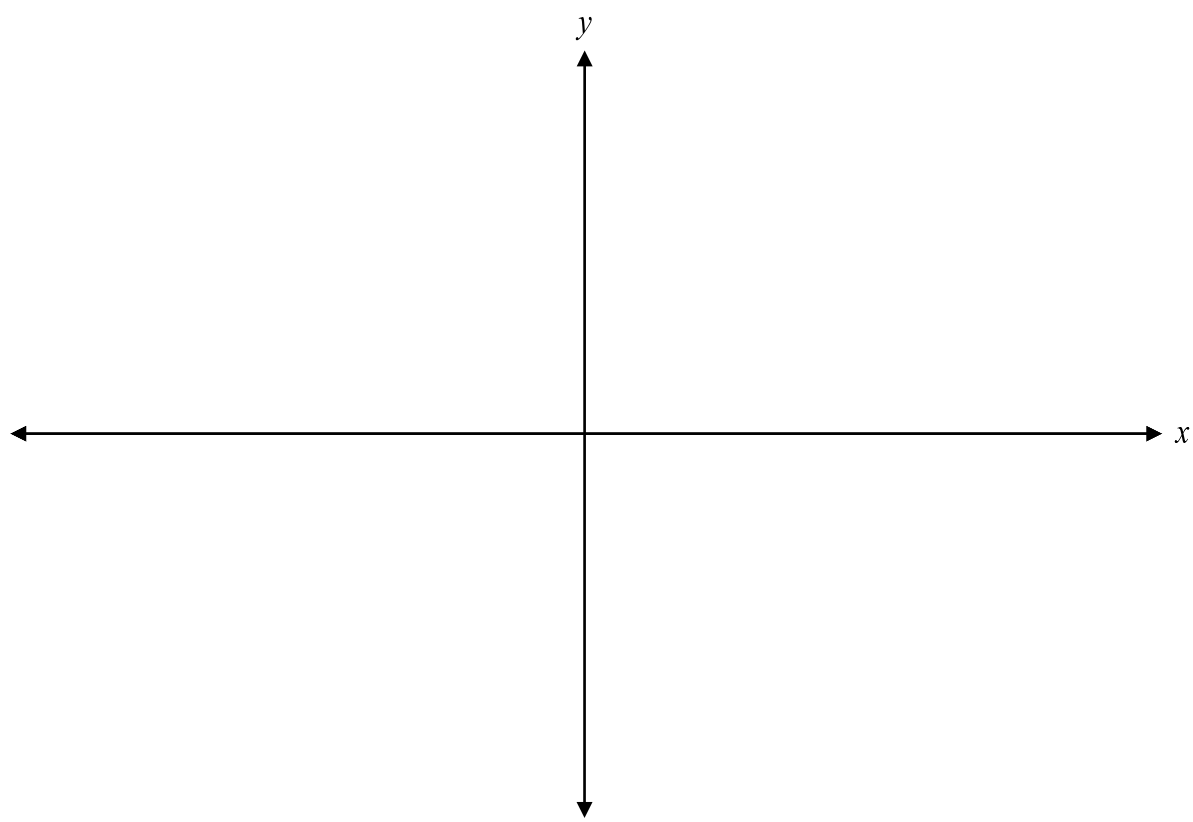
b) -1

c) 1

d) 16

Question 27 4 marks 119

Sketch the graph of at least one period of the function $y = -\cos\left(x + \frac{\pi}{4}\right) + 3$.



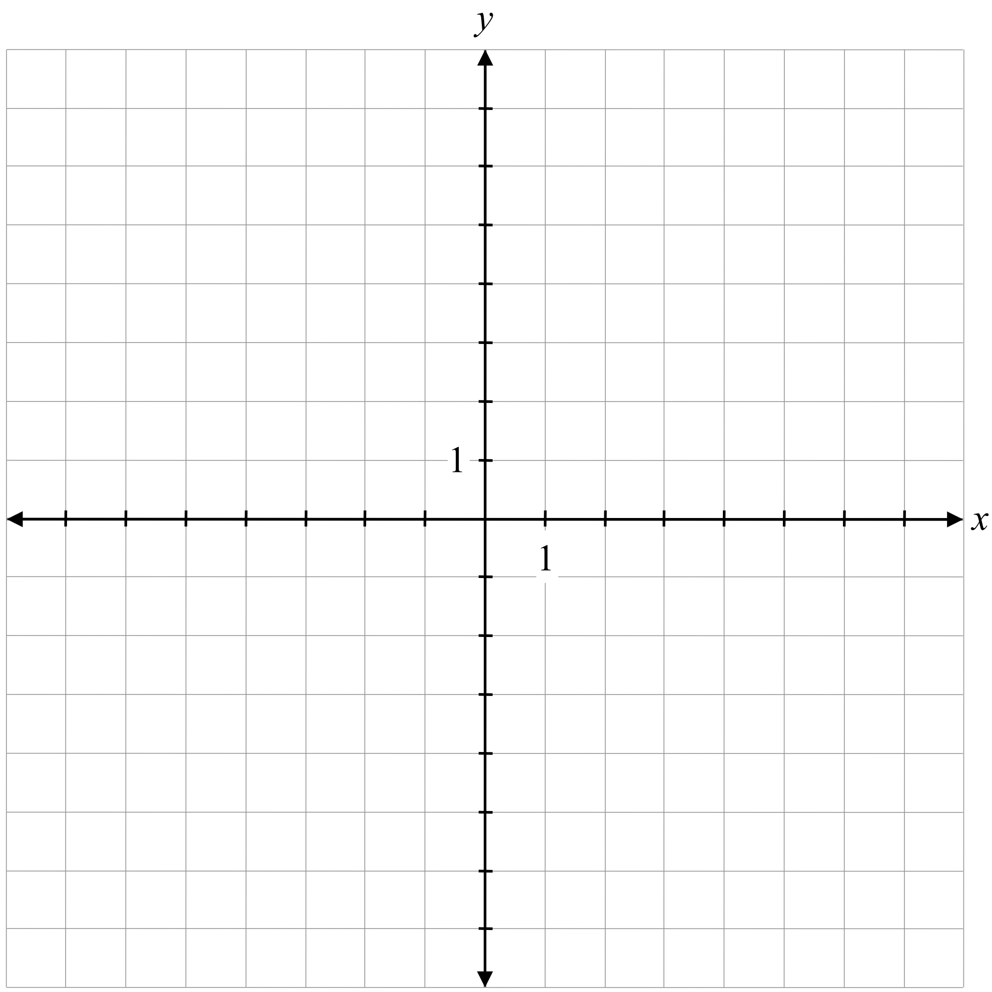
Question 28 1 mark 120

Justify that $(x-5)$ is not a possible factor of the function $P(x)=x^{3}-3x^{2}-4x+12$.

Question 29 4 marks

121

Sketch the graph of $f(x) = \dfrac{6}{(x+2)(x-3)}$ and state the y-intercept.



y-intercept: ______________________________

Question 30 2 marks 122

Determine how many 3-digit odd numbers less than 300 are possible using the digits 1, 2, 3, 4, 5, 6 if repetition is not allowed.

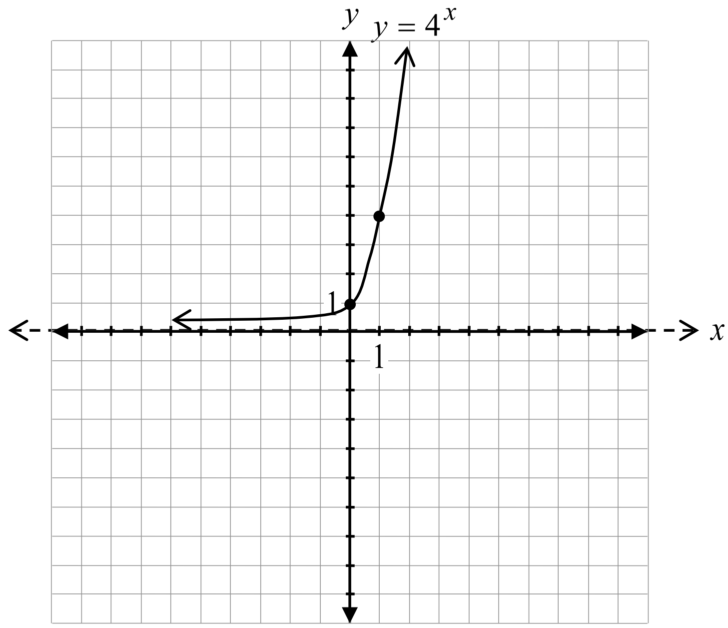
Question 31 3 marks

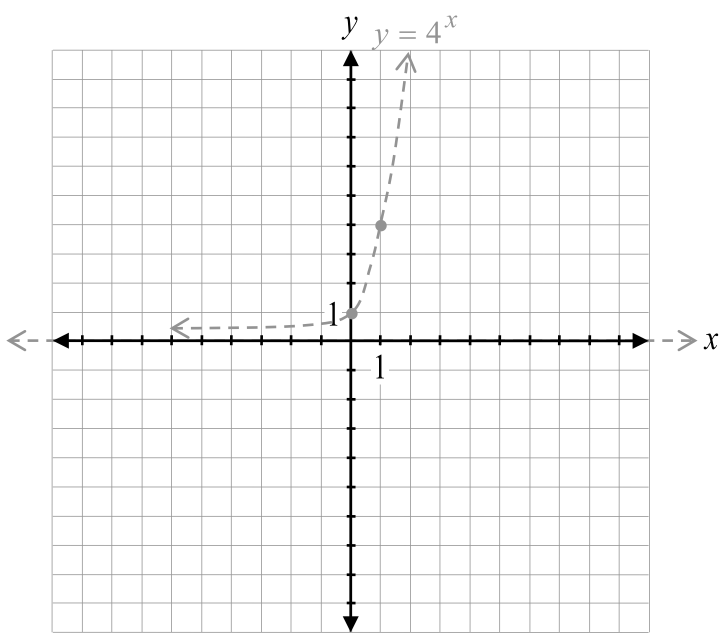
123

Given that $\cos\alpha = -\frac{5}{13}$ and $\sin\beta = \frac{2}{3}$, where α and β terminate in the same quadrant, determine the exact value of $\cos(\alpha - \beta)$.

Question 32 3 marks 124

Given the graph of $y = 4^x$, sketch the graph of $y = 2(4)^{x-3} + 1$.





The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 33 1 mark 125

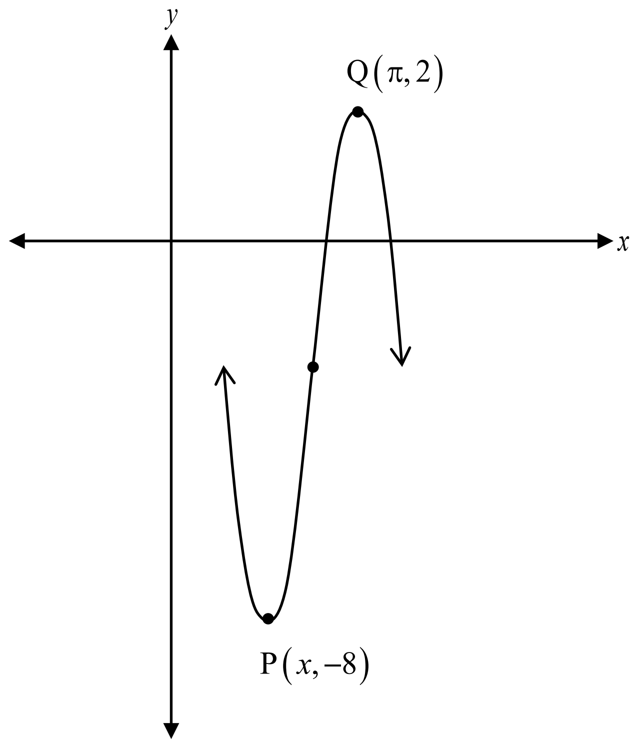
Determine the coterminal angle of $\frac{\pi}{5}$ over the interval $[-2\pi, 0]$.

Question 34 1 mark 126

State the domain of the graph of $y = \log(x-4) - 8$.

Question 35 1 mark

Given the graph of $y = 5\sin\left[2\left(x + \frac{\pi}{4}\right)\right] - 3$, determine the exact value of the x-coordinate in the point P.



Question 36 2 marks 128

Verify that the following equation is true for $x = \frac{5\pi}{6}$.

$$\frac{\cos x}{1 - \sin x} = \frac{1 + \sin x}{\cos x}$$

Left-Hand Side	Right-Hand Side

Question 37 2 marks

129

Given that $(x+1)$ is one of the factors of $P(x) = x^3 - x^2 + kx - 8$, determine the value of k.

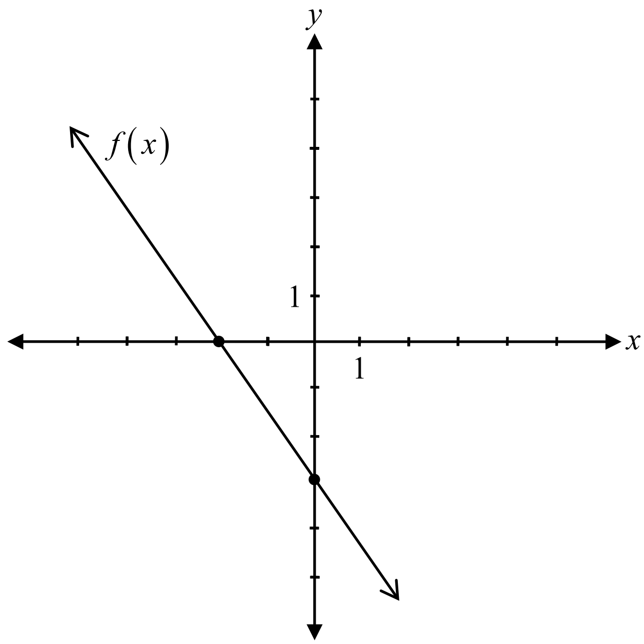
Question 38 — 1 mark

130

Given the function $f(x) = \sqrt{x}$, describe how to use transformations to determine the domain of the function $g(x) = f(x+2)+1$.

Question 39 — 1 mark

Given the graph of $y = f(x)$, state the equation of the vertical asymptote of $y = \dfrac{1}{f(x)}$.



Question 40 2 marks 132

Solve, algebraically.

$$16^x = 64^{2x-1}$$

Question 41 2 marks 133

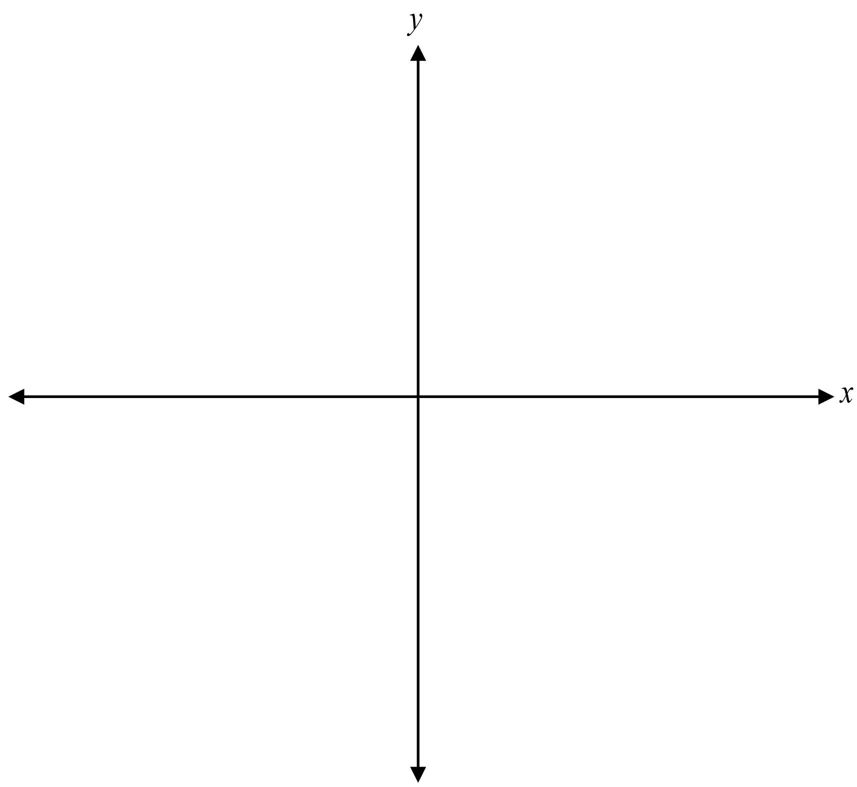
Given one of the factors of $P(x) = x^3 + 2x^2 - 5x - 6$ is $(x+3)$, express $P(x)$ in completely factored form.

$P(x) =$ ______________________________

Question 42 3 marks

134

Sketch the graph of $p(x) = 3(x+1)^2(x-2)^2$.

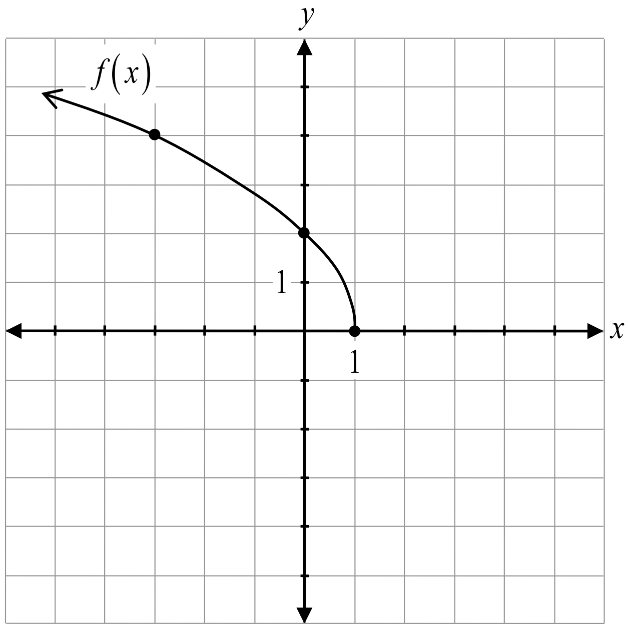


Question 43 2 marks 135

Given that $f(x) = x^2 - 4$ and $g(x) = \sqrt{x}$, determine $f(g(x))$ and state its domain.

$f(g(x)) =$ ______________________________

Determine a possible equation of the function $f(x)$.



$f(x) =$ ______________________________

Question 45 1 mark 137

Explain why the graph of $y = \log_2 x$ does not have a y-intercept.

Question 46 3 marks 138

Evaluate.

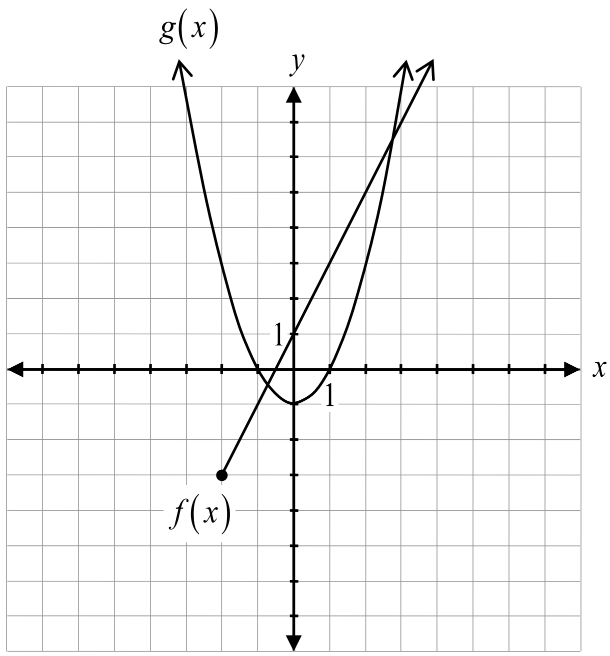
$$\sin^2\left(-\frac{\pi}{3}\right)+\cos\left(\frac{17\pi}{6}\right)\sec\left(\frac{\pi}{6}\right)$$

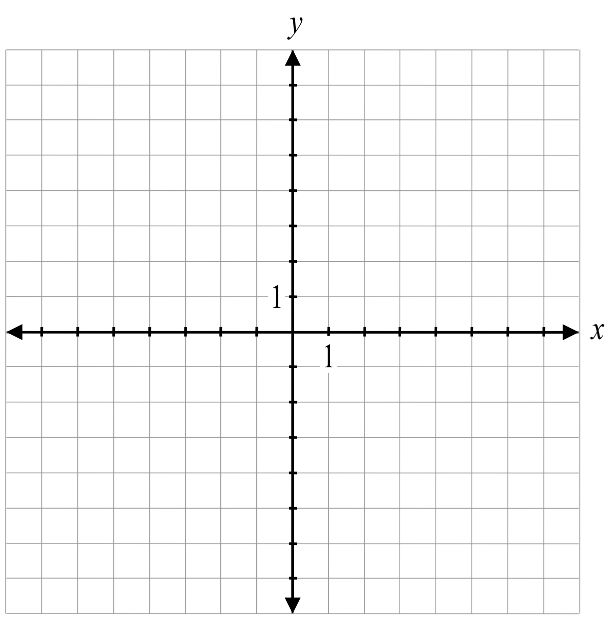
Question 47 1 mark 139

Determine the coordinates of the point of discontinuity (hole) on the graph of $y = \dfrac{x^2 - 3x}{x}$.

Question 48 — 2 marks 140

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $h(x) = f(x) + g(x)$.





Question 49 2 marks 141

Given that $\csc\theta = -\frac{4}{\sqrt{7}}$ and $\cos\theta > 0,$ determine the exact value of $\tan\theta.$

No marks will be awarded for work done on this page.